CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 2: Polynomials

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the zeroes of $x^2 + 9x + 14$ and verify the relationship between the zeroes and the coefficients.

2. Find the zeroes of $9x^2 - 12x + 4$.

3. Find the zeroes of $5x^2 - 20$.

4. Find a quadratic polynomial whose zeroes have sum 4 and product -5.

5. Find a quadratic polynomial whose zeroes are $\frac{2}{3}$ and $-\frac{1}{2}$.

6. If α and β are the zeroes of $x^2 - 6x + 8$, find $\alpha^2 + \beta^2$.

7. If α and β are the zeroes of $3x^2 + 5x - 2$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

8. If one zero of $x^2 - 8x + k$ is 6, find k and the other zero.

9. If the sum of the zeroes of $kx^2 + 2x + 3k$ is equal to their product, find k.

10. The zeroes of $x^3 - 4x^2 + x + 6$ are -1, 2 and 3. Verify the three relations between the zeroes and the coefficients.

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Answer Key

1. -2 and -7; sum $= -9$, product $= 14$
2. $\frac{2}{3}$ and $\frac{2}{3}$
3. 2 and -2
4. $x^2 - 4x - 5$
5. $6x^2 - x - 2$
6. $36 - 16 = 20$
7. $\frac{-5/3}{-2/3} = \frac{5}{2}$
8. $k = 12$; the other zero is 2
9. $-\frac{2}{k} = 3$, so $k = -\frac{2}{3}$
10. Sum $= 4$; sum of products in pairs $= -2 + 6 - 3 = 1$; product $= -6$

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